

Reinforcement Learning

SUBJECT CODE: 24DS742

Experiment 1: Exploring Reinforcement Learning Environments **Lab Task – 1**

Submitted by

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IN
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lab1.py

```
1 import gymnasium as gym
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 print("Gymnasium Version:", gym.__version__)
```

lab1_task2.py

```
1 import gymnasium as gym
2
3
4 env = gym.make("CartPole-v1")
5
6
7 observation, info = env.reset()
8
9 print("Initial Observation:")
10 print(observation)
11
12 print("\nEnvironment Info:")
13 print(info)
```

lab1_task3.py

```
1 import gymnasium as gym
2
3 env = gym.make("CartPole-v1")
4
5 observation, info = env.reset()
6
7 print("Initial Observation:")
8 print(observation)
9
10 print("\nEnvironment Info:")
11 print(info)
12
13 print("\nObservation Space:")
14 print(env.observation_space)
15
16 print("\nAction Space:")
17 print(env.action_space)
18
19 print("\nObservation Space Type:")
20 print(type(env.observation_space))
21
22 print("\nNumber of Possible Actions:")
23 print(env.action_space.n)
```

lab1_task4.py

```
1 import gymnasium as gym
2
3 # Create the CartPole environment
4 env = gym.make("CartPole-v1")
5
6 # Reset the environment
7 observation, info = env.reset()
8
9 # Initialize counters
10 step = 0
11 total_reward = 0
12
13 print("Starting Random Agent...\n")
14
15 # Run one episode
16 while True:
17
18     # Select a random action
19     action = env.action_space.sample()
20
21     # Execute the action
22     observation, reward, terminated, truncated, info = env.step(action)
23
24     # Update counters
25     step += 1
26     total_reward += reward
27
28     # Print step details
29     print(f"Step: {step}")
30     print(f"Action: {action}")
31     print(f"Observation: {observation}")
32     print(f"Reward: {reward}")
33     print(f"Terminated: {terminated}")
34     print(f"Truncated: {truncated}")
35     print("-" * 40)
36
37     # End the episode if finished
38     if terminated or truncated:
39         break
40
41 print("\nEpisode Finished!")
42 print(f"Total Steps: {step}")
43 print(f"Total Reward: {total_reward}")
44
45 # Close the environment
46 env.close()
```

RL_LAB

lab1_task2.py

lab1_task3.py

lab1_task4.py

lab1.ipynb

lab1.py

lab1.ipynb > M↓ Task - 2

Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ...

Python 3.9.6

Task - 1

markdown

[1]

```
import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt

print("Gymnasium Version:", gym.__version__)
```

Python

... Gymnasium Version: 1.1.1

Task - 2

markdown

[2]

```
import gymnasium as gym

env = gym.make("CartPole-v1")

observation, info = env.reset()

print("Initial Observation:")
print(observation)

print("\nEnvironment Info:")
print(info)
```

Python

... Initial Observation:

[0.00040509 0.01127484 -0.0328642 -0.03011545]

Environment Info:

{}

> OUTLINE

> TIMELINE

```
import gymnasium as gym

env = gym.make("CartPole-v1")

observation, info = env.reset()

print("Initial Observation:")
print(observation)

print("\nEnvironment Info:")
print(info)

print("\nObservation Space:")
print(env.observation_space)

print("\nAction Space:")
print(env.action_space)

print("\nObservation Space Type:")
print(type(env.observation_space))

print("\nNumber of Possible Actions:")
print(env.action_space.n)
```

[3]

Python

```
... Initial Observation:
[-0.0235572 -0.04018767 -0.04101782  0.04780161]
```

```
Environment Info:
{}
```

```
Observation Space:
Box([-4.8          -inf -0.41887903      -inf], [4.8          inf 0.41887903      inf], (4,), float32)
```

```
Action Space:
Discrete(2)
```

Observation Space Type:
<class 'gymnasium.spaces.box.Box'>

Number of Possible Actions:
2

Task – 4

markdown

```
import gymnasium as gym

# Create the CartPole environment
env = gym.make("CartPole-v1")

# Reset the environment
observation, info = env.reset()

# Initialize counters
step = 0
total_reward = 0

print("Starting Random Agent...\n")

# Run one episode
while True:

    # Select a random action
    action = env.action_space.sample()

    # Execute the action
    observation, reward, terminated, truncated, info = env.step(action)

    # Update counters
    step += 1
    total_reward += reward

    # Print step details
    print(f"Step: {step}")
    print(f"Action: {action}")
```




```
# End the episode if finished
if terminated or truncated:
    break

print("\nEpisode Finished!")
print(f"Total Steps: {step}")
print(f"Total Reward: {total_reward}")

# Close the environment
env.close()
```

[4]

Python

... Starting Random Agent...

```
Step: 1
Action: 1
Observation: [-0.04828697  0.18795209 -0.04118809 -0.27073267]
Reward: 1.0
Terminated: False
Truncated: False
```

```
-----
Step: 2
Action: 0
Observation: [-0.04452793 -0.00655865 -0.04660275  0.00868011]
Reward: 1.0
Terminated: False
Truncated: False
```

```
-----
Step: 3
Action: 1
Observation: [-0.0446591  0.1891996 -0.04642915 -0.29833472]
Reward: 1.0
Terminated: False
Truncated: False
```

```
-----
Step: 4
Action: 1
```

...

```
Episode Finished!
Total Steps: 12
Total Reward: 12.0
```

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